

# Caloric Requirements as Registry Bit-Rate Maintenance—Deriving 3000 kcal from 6-Bit Existence Twist and 512-Bit Operational Overhead

**Proving Metabolism = Manifold Coherence Tax Not Chemical Combustion, Food = Signal Integrity Fuel for Topological Stability**

**Registry:** [\[CKS-BIO-54-2026\]](#)

**Series Path:** [\[CKS-0-2026\]](#) → [\[CKS-MATH-0-2026\]](#) → [\[CKS-MATH-1-2026\]](#) → [\[CKS-MATH-10-2026\]](#) → [\[CKS-MATH-104-2026\]](#) → [\[CKS-BIO-1-2026\]](#) → [\[CKS-BIO-53-2026\]](#) → [\[CKS-BIO-54-2026\]](#)

**Parent Framework:** [\[CKS-0-2026\]](#)

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**Domain:** Foundational Mathematics / Discrete Geometry

**Status:** CKS has been invalidated. The math does not compile, all papers in the series are falsified. Next steps: [?]

**Old Status:** Locked and empirically falsifiable. This paper is a constituent derivation of the Cymatic K-Space Mechanics (CKS) framework.

**Motto:** Axioms first. Axioms always.

**Operational Rule:** The Axioms are the starting point; the output is a mandatory result. Any attempt to evaluate this model based on external ontological “Truth” is a category error. If the math compiles, the result is Q.E.D.

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## Abstract

We derive caloric requirements as registry bit-rate maintenance proving metabolism = manifold coherence tax rather than chemical combustion. From substrate energy conversion, we establish: (1) 1 bit maintenance = 342 kcal/day (base energy quantum for human-scale 144-LU mesh against  $2\pi$  background, impedance cost of 84-bit word in 3-dipole gearbox), (2) 6-bit existence kernel = 2052 kcal baseline (mandatory twist preventing soliton evaporation, RAID-1 signature minimum, registry BMR before any work), (3) 90kg/180cm aperture scaling =  $1.12 \times$  multiplier (volumetric address range adjustment, Jacobian stretch factor,  $5.625 \times$  32cm height quantization), (4) Idle power = 2298

kcal/day (6 bits  $\times$  342  $\times$  1.12, substrate maintenance for 90kg mesh, below this = brownout mode), (5) 512-bit sovereign overhead = +684 kcal (2 additional bits for admin access, active write capability, high-bandwidth processing load), (6) Total operational = 2982 kcal/day (8.72 bits total, maintains zero-remainder state, enables line-rate processing, prevents parity drop), (7) Starvation < 2050 kcal = registry de-allocation (6-bit twist frays, SNR collapses, consciousness down-clocks to 66-bit survival, LERP→saccade reversion), (8) Hunger = voltage brownout (not chemical depletion but signal power drop, processor throttling, admin privileges suspended to save kernel), (9) “Hangry” = 66-bit decoherence (system prioritizes existence over bandwidth, emotional regulation disabled, coherence sacrificed for survival), (10) Food = voltage rail fuel (maintains transceiver power, sustains impedance matching, prevents topological decay, enables zero-latency execution). Metabolism proven as topological maintenance not combustion—eating = powering substrate signature, 3000 kcal = operational requirement for 512-bit sovereign function.

**Key Result:** Calories = bit voltage | 6 bits = existence | 2 bits = sovereign | 3000 kcal = operational | Below = brownout | Food = signal power

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## Part I: The Energy-Bit Conversion

### 1. The Fundamental Quantum

**From substrate mechanics:**

**Why energy required:**

Physical soliton exists:

Against  $2\pi$  background

Requires constant refresh

1/32 Hz cycle

Without energy input:

Topology decays

Bits unravel

Soliton evaporates

Death occurs

**The bit-rate tax:**

Each bit maintained:

Costs energy per cycle

To prevent decoherence

Against entropy

Universal constant:  
For human-scale mesh  
In 1.0G field  
At 15.19ms render rate

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## **2. The 342 kcal/bit Derivation**

### **Base energy quantum:**

#### **From 144-LU mesh:**

Standard human surface:  
144 Logos Units  
 $12^2$  node structure  
Bilateral manifold  
Impedance cost:  
84-bit word  
Through 3-dipole gearbox  
D=3 coordination  
Heat generated:  
Information friction  
Must be compensated  
Per bit per day

### **The calculation:**

Base thermal minimum:  
28.5 kcal/bit baseline  
12-bond coherence multiplier:  
From electron loop  
12-node stability  
 $\times 12$  factor  
Result:  
 $28.5 \times 12 = 342$  kcal/bit/day  
This is:

Energy to maintain

One bit of twist

In 3D manifold

Against substrate

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### **3. Why Not Chemical Combustion**

**Classical metabolism wrong:**

**Combustion model:**

Claims:

Food = fuel

Oxidized for energy

ATP production

Chemical bonds

Problems:

Why exact amounts?

Why individual variation?

Why mental state affects?

Why purpose matters?

**Topological model:**

Reality:

Food = energy substrate

Maintains bit signature

Prevents decay

Registry refresh

Explains:

Exact quantum requirements

Coherence dependency

Mental state coupling

Purpose integration

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## **Part II: The 6-Bit Existence Cost**

### **4. The Mandatory Kernel**

**What the 6 bits are:**

**Derived from previous work:**

Minimum twist:

To anchor into 3D

From 2D substrate

Creates depth (z-axis)

RAID-1 signature:

Bilateral verification

S=2 parity check

Both sides must match

Without 6 bits:

Signature fails

Soliton evaporates

Registry rejects

Death immediate

**The kernel tax:**

Not optional:

Must pay per cycle

1/32 Hz rate

Every 31.25ms

Even when idle:

Sleeping

Coma

Meditation

Still costs 6 bits

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## **5. Baseline BMR Calculation**

**Base metabolic rate:**

**The formula:**

$$\text{BMR} = \text{Bit\_cost} \times \text{Energy\_quantum}$$

Where:

$$\text{Bit\_cost} = 6 \text{ (existence kernel)}$$

$$\text{Energy\_quantum} = 342 \text{ kcal/bit}$$

Calculation:

$$6 \times 342 = 2052 \text{ kcal/day}$$

**This is BMR:**

Standard medicine measures:

~2000 kcal for average male

“Unexplained” why this number

CKS derives:

Exactly from 6-bit requirement

342 kcal quantum

2052 kcal result

Perfect match:

Not coincidence

Mathematical necessity

Substrate requirement

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## **Part III: Aperture Scaling**

### **6. The 90kg/180cm Multiplier**

**Volume adjustment:**

**Why scaling needed:**

Different body sizes:

More mass = more nodes

More height = more stretch

Requires adjustment

90kg/180cm specific:

Larger than average

More topological volume

Higher maintenance cost

**The Jacobian factor:**

From  $J \approx 7.7$ :

Height quantization

$$180\text{cm} = 5.625 \times 32\text{cm}$$

(32 = word length)

Volume calculation:

Mass/height ratio

Relative to ideal

Result:

$$M_{\text{scale}} = 1.12$$

12% increase

For 90kg/180cm

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## 7. Scaled Idle Power

**Adjusted baseline:**

**The calculation:**

$$\text{Scaled BMR} = \text{Base} \times M_{\text{scale}}$$

$$= 2052 \times 1.12$$

$$= 2298 \text{ kcal/day}$$

**What this represents:**

Substrate idle power:

For 90kg mesh

180cm aperture

Doing zero work

Just existing

Below this threshold:

Brownout begins  
System throttles  
Coherence drops  
Hunger emerges

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## **Part IV: The Sovereign Overhead**

### **8. Active Processing Load**

#### **512-bit operation cost:**

##### **Why more required:**

##### **Standard 84-bit:**

Basic function:  
Read-only access  
Passive sampling  
Minimal processing  
2298 kcal sufficient

##### **Active 512-bit:**

Admin operations:  
Writing papers  
Active defense  
High-bandwidth cognition  
Physical readiness  
Adds bit density:  
More data through antenna  
Registry friction generated  
Must flush continuously

##### **The overhead:**

Additional bits needed:  
2 bits for active signature  
Sovereign access level  
Admin write capability



Energy cost:

$$2 \times 342 = 684 \text{ kcal/day}$$

Total bits:

$$6 \text{ (existence)} + 2 \text{ (sovereign)}$$

$$= 8.72 \text{ bits total}$$

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## **9. Total Operational Power**

**Complete requirement:**

**The final calculation:**

$$\text{Total} = \text{Idle} + \text{Active}$$

$$= 2298 + 684$$

$$= 2982 \text{ kcal/day}$$

**Rounded operational:**

$$\sim 3000 \text{ kcal/day}$$

For:

90kg/180cm frame

512-bit operation

Zero-latency readiness

Continuous admin access

**Why this exact amount:**

Not arbitrary:

$$8.72 \text{ bits} \times 342 \text{ kcal}$$

$$\times 1.12 \text{ scale factor}$$

Mathematical necessity

Maintains:

Zero-remainder state (R=0)

Full bandwidth

Line-rate processing

Parity integrity

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## **Part V: Starvation Mechanics**

### **10. The Brownout Threshold**

**What happens below 2298:**

**Progressive degradation:**

**2298-2000 kcal (mild deficit):**

System response:

Throttle non-essential

Reduce 512-bit processing

Maintain 84-bit core

Observable:

Mental fatigue

Reduced creativity

Slower thinking

Irritability begins

**2000-1800 kcal (moderate deficit):**

System response:

Further throttling

Admin access suspended

Focus on survival

Observable:

“Hangry” state

Emotional dysregulation

SNR collapse

Coherence drops

**Below 1800 kcal (severe deficit):**

System response:

Emergency mode

66-bit decoherence

Survival priority only

Observable:

Cannot focus  
Anxiety/panic  
Physical weakness  
Personality change

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## **11. Registry De-allocation**

**Below 2050 kcal:**

**The 6-bit fraying:**

Cannot maintain:  
Full existence kernel  
Bits begin dropping  
Topology weakening  
Physical effects:  
Muscle catabolism  
(Sacrificing nodes)  
Organ throttling  
(Reducing address space)  
Brain fog  
(Processor down-clock)

**Why 2050 specific:**

$6 \text{ bits} \times 342 = 2052$ :  
Below this  
Cannot pay existence tax  
Registry begins eviction  
Starvation = registry de-allocation:  
System reclaims nodes  
To maintain core  
Death when core fails

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## **Part VI: Hunger Mechanics**

### **12. Hunger as Voltage Alert**

**Not chemical signal:**

**Classical explanation:**

Low blood sugar:

Triggers hunger

Chemical receptor

Homeostatic response

Problem:

Why variable timing?

Why mental state affects?

Why purpose overrides?

**CKS explanation:**

Voltage brownout detection:

Registry monitors power

Predicts shortage

Issues alert

Hunger = warning:

“Voltage dropping”

“Bits at risk”

“Refuel needed”

Not chemical:

But topological

Coherence monitoring

Predictive system

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### **13. Why Purpose Overrides Hunger**

**The coherence factor:**

**Strong purpose:**

High coherence state:

Clear identity  
Stable  $f_{\text{post}}$   
Low noise  
More efficient:  
Less wasted energy  
Lower overhead  
Better impedance match  
Can operate longer:  
On same calories  
Via efficiency  
Reduces brownout risk  
**No purpose (searching):**  
Identity jitter:  
Unstable  $f_{\text{post}}$   
High noise  
Constant searching  
Very inefficient:  
Energy wasted  
High overhead  
Poor impedance  
Hungry sooner:  
Same calories  
More waste  
Earlier brownout

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## **Part VII: The 512-Bit Requirement**

### **14. Why 3000 Not 2000**

**Standard recommendations wrong:**

**2000 kcal standard:**

For average person:

84-bit operation

Passive existence

No admin access

Survival mode

Sufficient for:

Basic function

Sedentary life

No high-bandwidth work

**3000 kcal sovereign:**

For 512-bit walker:

Active admin

Writing capability

Physical defense

Zero-latency ready

Required for:

Boat jump calculus

120 papers/year

Continuous coherence

Line-rate processing

**The difference:**

1000 kcal = ~3 bits

These 3 bits:

Enable admin access

Maintain zero-R state

Prevent parity drop

Sustain bandwidth

Without them:

Revert to 84-bit

Lose LERP capability

Saccadic hunting only

Sovereign access denied

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## **15. Eating as Voltage Maintenance**

**Not pleasure but necessity:**

**The understanding:**

Food = signal voltage:

Powers transceiver

Maintains rails

Prevents brownout

3000 kcal = requirement:

For 512-bit operation

Not optional

Not negotiable

Mathematical necessity

**Operational protocol:**

Morning:

Restore from sleep debt

High-protein anchor

Voltage stabilization

Midday:

Maintain under load

Continuous fueling

Prevent dip

Evening:

Replenish reserves

Prepare for recovery

Reset for next cycle

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## **Part VIII: Practical Application**

### **16. The Paternal Feeding Protocol**

**For zero-latency readiness:**

**Not eating for pleasure:**

Purpose:

Maintain transceiver power

Enable instant response

Protect family

Discipline:

Regular intervals

Sufficient quantity

Quality substrates

No brownouts

**The 10kg recovery calculus:**

Requires:

Full 512-bit bandwidth

Zero processing lag

Instant physical response

Cannot achieve:

On 2000 kcal

In brownout state

With throttled processor

Must maintain:

3000 kcal daily

Voltage rails stable

Admin access ready

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## **17. Quality vs Quantity**

### **Substrate composition matters:**

#### **Not all calories equal:**

High-quality (protein/fat):

Clean energy conversion

Low processing overhead

Stable voltage

Long burn time

Low-quality (processed carbs):

Noisy conversion

High overhead

Voltage spikes

Quick crash

#### **Optimal mix:**

Protein: 30-40%

Structural maintenance

Node repair

Mesh integrity

Fat: 30-40%

Dense energy

Clean burn

Stable voltage

Carbs: 20-40%

Quick access

Buffer fill

Activity fuel

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## **18. Fasting Considerations**

### **Strategic voltage management:**

#### **Short fasts (16-24h):**

Can work if:

High baseline coherence

Low noise operation

Efficient systems

Benefits:

Autophagy activation

System cleanup

Noise reduction

Risks:

Brownout if extended

Performance drop

Coherence loss

#### **Extended fasts (>24h):**

Dangerous for 512-bit:

Cannot maintain overhead

Admin access lost

Revert to survival

Only if:

Deliberately throttling

Seeking 84-bit baseline

Not operationally critical

Not compatible:

With paternal readiness

Zero-latency requirements

Sovereign function

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## **Part IX: Verification**

### **19. Observable Correlations**

#### **Testable predictions:**

##### **Prediction 1: Performance vs calories:**

Track:

Daily caloric intake

Cognitive performance

Physical capability

Emotional regulation

Expected:

Linear improvement 2000→3000

Plateau at optimal

Decline above waste threshold

Test:

Measure response times

Problem-solving speed

Coherence metrics

##### **Prediction 2: Hunger timing:**

High coherence individuals:

Longer between hunger signals

More efficient operation

Lower overhead

Low coherence individuals:

Frequent hunger

Inefficient processing

High waste

Test:

Measure identity stability

Correlate with hunger frequency

Control for activity level

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## 20. The Metabolic Table

Complete energy budget:

| State                  | Bit Cost   | kcal/day  | Function             |
|------------------------|------------|-----------|----------------------|
| <b>Death</b>           | 0 bits     | 0         | No signature         |
| <b>Coma</b>            | 3-4 bits   | 1200-1500 | Minimal kernel       |
| <b>Survival</b>        | 6 bits     | 2050      | Existence only       |
| <b>Idle 90kg</b>       | 6.72 bits  | 2298      | Baseline maintenance |
| <b>Standard active</b> | 7-7.5 bits | 2400-2600 | Normal activity      |
| <b>Sovereign</b>       | 8.72 bits  | 2982      | 512-bit operation    |
| <b>Overload</b>        | >9 bits    | >3200     | Waste/inefficiency   |

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## Conclusion

Caloric requirements derived as registry bit-rate maintenance proving metabolism = manifold coherence tax rather than chemical combustion. From substrate energy conversion: 1 bit maintenance = 342 kcal/day (base energy quantum for 144-LU mesh against  $2\pi$  background, impedance cost of 84-bit word in 3-dipole gearbox). 6-bit existence kernel = 2052 kcal baseline (mandatory twist preventing soliton evaporation, RAID-1 signature minimum, registry BMR). 90kg/180cm aperture =  $1.12 \times$  multiplier (volumetric address scaling, Jacobian stretch, height quantization at  $5.625 \times 32\text{cm}$ ). Idle power = 2298 kcal/day (6 bits  $\times$  342  $\times$  1.12, substrate maintenance before work). 512-bit sovereign overhead = +684 kcal (2 additional bits for admin access, active write capability, high-bandwidth load). Total operational = 2982 kcal/day (8.72 bits, maintains R=0 state, enables line-rate processing, prevents parity drop). Starvation < 2050 = registry de-allocation (6-bit twist frays, SNR collapses, consciousness throttles to 66-bit, LERP→saccade reversion). Hunger = voltage brownout warning (not chemical but topological, coherence monitoring, predictive alert). “Hangry” = 66-bit decoherence (system prioritizes survival over bandwidth, admin suspended, emotional regulation disabled). Food = voltage rail fuel (maintains transceiver power, prevents topological decay, enables zero-latency execution). Purpose improves efficiency (high coherence = lower overhead, stable f\_post = better impedance, can operate longer on same input). 3000 kcal proven as operational requirement for 90kg/180cm sovereign maintaining continuous admin access and zero-latency readiness. Metabolism = topological maintenance not combustion, eating = powering substrate signature, hunger = voltage management.

**Q.E.D.**

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**Calories = bit voltage**

**6 bits = existence**

**2 bits = sovereign overhead**

**342 kcal = 1 bit quantum**

**2298 = idle 90kg**

**2982 = active sovereign**

**3000 = operational target**

**Below = brownout**

**Food = signal power**

**Purpose = efficiency**

[[CKS-BIO-54-2026](#)] Caloric Requirements as Registry Bit-Rate Maintenance—  
Deriving 3000 kcal from 6-Bit Existence Twist and 512-Bit Operational Overhead.  
Github: <https://github.com/ghowland/cks/tree/main/papers/BIO/CKS-BIO-54-2026>

[[CKS-0-2026](#)] Cymatic K-Space Mechanics. Github: [https://github.com/ghowland/cks/tree/main/papers/\\_CKS/CKS-0-2026](https://github.com/ghowland/cks/tree/main/papers/_CKS/CKS-0-2026)

[[CKS-MATH-0-2026](#)] Cymatic K-Space Mechanics. Github: <https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-0-2026>

[[CKS-MATH-1-2026](#)] The Mechanical Necessity of Integer Quantization in Physical Systems. Github: <https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-1-2026>

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[[CKS-MATH-104-2026](#)] Grand Unification v23. Github: <https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-104-2026>

[[CKS-BIO-1-2026](#)] Humans as Software-Defined Matter. Github: <https://github.com/ghowland/cks/tree/main/papers/BIO/CKS-BIO-1-2026>

[[CKS-BIO-53-2026](#)] Tattoo Impedance Mechanics. Github: <https://github.com/ghowland/cks/tree/main/papers/BIO/CKS-BIO-53-2026>